That's increasingly no longer true. Many technologies and systems are becoming so complex that they're beyond the capacity of any one individual to truly understand them … In AI, the neural networks moving toward autonomy are, at present, not explainable. You can't walk someone through the decision-making process to explain precisely why an algorithm produced a specific prediction. Engineers can't peer beneath the hood and easily explain in granular detail what caused something to happen. GPT-4, AlphaGo, and the rest are black boxes, their outputs and decisions based on opaque and impossibly intricate chains of minute signals.'[36]

The rise of unfathomable alien intelligence undermines democracy. If more and more decisions about people's lives are made in a black box, so voters cannot understand and challenge them, democracy ceases to function. In particular, what happens when crucial decisions not just about individual lives but even about collective matters like the Federal Reserve's interest rate are made by unfathomable algorithms? Human voters may keep choosing a human president, but wouldn't this be just an empty ceremony? Even today, only a small fraction of humanity truly understands the financial system. A 2016 survey by the OECD found that most people had difficulty grasping even simple financial concepts like compound interest.[37] A 2014 survey of British MPs – charged with regulating one of the world's most important financial hubs – found that only 12 per cent accurately understood that new money is created when banks make loans. This fact is among the most basic principles of the modern financial system.[38] As the 2007–8 financial crisis indicated, more complex financial devices and principles, like those behind CDOs, were intelligible to only a few financial wizards. What happens to democracy when AIs create even more complex financial devices and when the number of humans who understand the financial system drops to zero?

The increasing unfathomability of our information network is one of the reasons for the recent wave of populist parties and charismatic leaders. When people can no longer make sense of the world, and when they feel overwhelmed by immense amounts of information they cannot digest, they become easy prey for conspiracy theories, and they turn for salvation to something they do understand – a human. Unfortunately, while charismatic leaders certainly have their advantages, no single human, however inspiring or brilliant, can single-handedly decipher how the algorithms that increasingly dominate the world work, and make sure that they are fair. The